



# ChemBuddy

The Intelligent Academic Operating System for MSc Chemistry Students

Version 3.5.1 • Complete Visual Operating Manual

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**Target Curriculum:** MSc Organic, Inorganic, Physical & Analytical Chemistry

**University Alignment:** BCU, Central Universities & UGC Model Syllabus

**Platform:** Android (Flutter SDK • 100% Offline Core • Gemini AI)

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## 1. Introduction & Academic Architecture

**ChemBuddy** is an advanced, offline-first mobile application designed specifically to eliminate the academic friction experienced by Master of Science (MSc) Chemistry postgraduates. Developed by **Prajwal A Kambar**, it merges rigorous chemoinformatics, high-precision spectroscopy solvers, attendance predictive mathematics, and state-of-the-art generative AI into a unified laboratory and classroom companion.

### Core Engineering Principles

- **100% Offline Core:** Attendance calculations, timetable intelligence, 16 postgrad calculators, Woodward-Hoffmann pericyclic selection rules, spectroscopy databases, and reaction prediction algorithms function

entirely on-device without internet connectivity.

- **Scientific Rigor:** All formulas, mechanisms, spectroscopic chemical shifts, and thermodynamic equations strictly adhere to IUPAC standards and standard graduate textbooks (Silverstein, Pavia, March, Clayden).
- **Student Privacy & Local Ownership:** Student profiles, attendance logs, notes, and uploaded PDFs reside securely in encrypted local Hive boxes on your device. Zero telemetry or surveillance.
- **Dark Laboratory Ergonomics:** The UI utilizes high-contrast deep purple ( #0A0914 ) and cyan ( #06B6D4 ) palettes engineered to preserve dark-adapted vision in laser labs and darkrooms.

## 2. Installation, Android Permissions & Initial Setup

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1. **Download Release APK:** Download `ChemBuddy-v3.5.1-by-Prajwal-A-Kambar.apk` from the official GitHub Releases page.
2. **Enable Unknown Sources:** In Android Settings, navigate to *Security* → *Install Unknown Apps* and allow your browser or file manager to install APKs.
3. **Grant Essential Permissions:**
  - **Notifications:** Required for the 8:30 AM Timetable Briefing, 30-min class alerts, and 8:00 PM Psychology Facts.
  - **Alarms & Reminders (Android 12+):** Required so that scheduled class notifications fire with exact timing.
  - **Battery Optimization:** Set to "*Unrestricted*" so Android Doze does not suppress morning notifications.
4. **Profile Initialization:** On first launch, enter your Name, College/University, Semester, and Specialization (Organic, Inorganic, Physical, or Analytical).

### 3. 🌟 Ask ChemBuddy AI — The Chemistry Intelligence Engine (FLAGSHIP FEATURE)

#### 💡 Flagship Feature Spotlight: ChemBuddy AI

**ChemBuddy AI** is your 24/7 MSc Chemistry Research & Exam Companion. Accessible via the **Ask AI** bottom navigation tab or the Home Screen action card, it connects directly to Google Gemini models (with intelligent offline fallbacks) and provides 9 specialized academic modes, chemistry-aware speech recognition, and instant active recall flashcard generation.

#### 3.1 Overview of the ChemBuddy AI Interface

The Ask ChemBuddy screen is designed to behave like an interactive academic interview with a senior chemistry professor. Key components include:

- **Horizontal Mode Scroller:** Tap between 9 specialized AI modes instantly.
- **Chemistry Speech-to-Text:** Tap the pulsing purple microphone button to dictate questions verbally. ChemBuddy automatically normalizes spoken phrases (e.g. converting "SN1" to  $S_N1$ , "Huckel rule" to  $4n+2$ , and "H-NMR" to  $^1H\text{ NMR}$ ).
- **KaTeX Equation Box:** Mathematical and chemical expressions are formatted in crisp, textbook LaTeX notation.
- **Electron Pushing & Curved Arrows:** Step-by-step mechanism breakdowns detail electron displacement from nucleophile to electrophile.
- **1-Tap Flashcard Generation:** Save any AI response directly into your spaced repetition flashcard deck with a single tap.

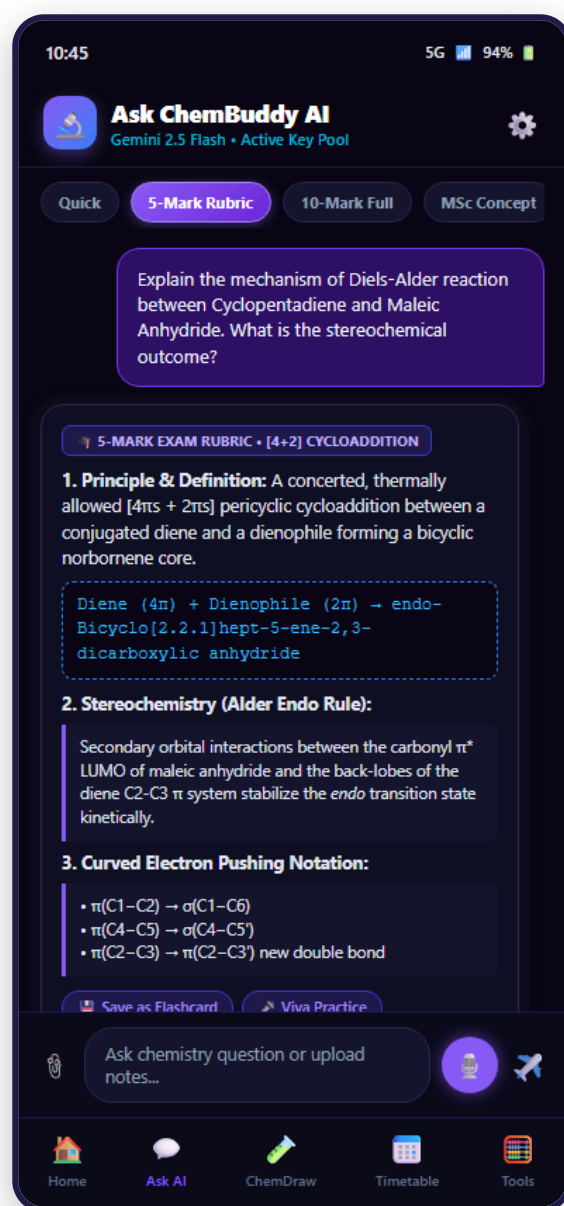


Figure 3.1: Ask ChemBuddy AI Interface with 5-Mark Exam Rubric mode, KaTeX math, and step-by-step electron pushing notation.

## 3.2 The 9 Specialized AI Modes Explained

| Mode Name                 | Target Output                    | Best Used For                                                                                                    |
|---------------------------|----------------------------------|------------------------------------------------------------------------------------------------------------------|
| 1. Quick Answer           | High-yield, concise fact         | Quick lab lookups, boiling points, reagents, oxidation states, and formulas.                                     |
| 2. General AI             | Broad generative reasoning       | Open-ended chemistry discussions, general scientific writing, and multi-turn research inquiries.                 |
| 3. Simple Explanation     | ELI5, jargon-free analogies      | Breaking down intimidating concepts (e.g., Berry pseudorotation, NMR relaxation times) into plain English.       |
| 4. 2-Mark University Exam | Concise, high-density definition | Direct definitions, IUPAC rules, and single-equation university exam questions.                                  |
| 5. 5-Mark Exam Rubric     | Structured 5-section essay       | Standard university exam questions: 1) Principle, 2) Balanced Reaction, 3) Mechanism, 4) Scope, 5) Summary.      |
| 6. 10-Mark Full Rubric    | Comprehensive long-form essay    | Major university questions with curved-arrow pushing, orbital transition states, and stereochemical proofs.      |
| 7. MSc Concept Deep-Dive  | Orbital & thermodynamic theory   | Frontier Molecular Orbital (FMO) analysis, perturbation theory, and physical organic chemistry intuition.        |
| 8. Stepwise Mechanisms    | Curved arrow electron flow       | Detailed step-by-step reaction mechanisms, nucleophilic attacks, carbocation rearrangements, and driving forces. |
| 9. From My PDF (RAG)      | Strictly grounded in notes       | Answers locked 100% to your uploaded syllabus, class notes, or research papers. Zero hallucinations.             |

## 3.3 Chemistry Speech-to-Text Normalization

When dictating via the microphone button (  ), ChemBuddy passes speech through the proprietary `ChemistryTextFormatter.normalizeSpeechQuery()` engine. Common spoken terms are converted to proper scientific formatting:

```
"s n 1 reaction" → "S_N1 reaction"
"huckel 4 n plus 2 rule" → "Hückel (4n+2) π-electron rule"
"proton n m r splitting" → "1H NMR splitting pattern"
"diels alder reaction" → "Diels-Alder [4+2] cycloaddition"
```

## 3.4 Multi-Key Gemini AI Orchestrator

To guarantee 99.9% uptime and prevent Google Gemini rate limits (HTTP 429), ChemBuddy utilizes an intelligent key-pool orchestrator:

- **Round-Robin Key Rotation:** Distributes requests across multiple API keys.

- **Automatic Failover:** If a key reaches its quota, the app automatically fails over to the next key without alerting the user.
- **Model Fallback Hierarchy:** Automatically tries `gemini-2.5-flash` → `gemini-2.0-flash` → `gemini-1.5-flash`.
- **Offline Synthesizer:** If completely offline, the local reaction engine and textbook databases take over seamlessly.

## 4. Home Dashboard & Dynamic Contextual Greetings

The **Home Dashboard** serves as the central command center of ChemBuddy. It gives you an immediate overview of your academic health, daily schedule, and cognitive enrichment upon opening the app.

### 4.1 Claude-Inspired Contextual Greetings

The header dynamically adapts based on time of day, day of the week, and your real-time attendance percentage:

- **Late Night (11 PM - 4 AM):** "Burning the midnight Bunsen burner, Prajwal? Don't forget sleep consolidates memory."
- **Morning (5 AM - 12 PM):** "Good morning, Prajwal 🧪  
Catalysis is chemistry with a shortcut."
- **Attendance Warning (<75%):** "⚠️ Attendance critical (72.4%). You cannot afford to miss today's lectures!"
- **Excellence Praise (>85%):** "🌟 Outstanding attendance health (91.2%). Safe bunks available: 5 classes."

### 4.2 Attendance Health Ring

A circular progress ring visually communicates your university compliance status:

- **Green ( $\geq 75\%$ ):** Exam Eligible. Displays the exact number of classes you can safely skip.
- **Red ( $< 75\%$ ):** Attendance Shortage. Displays the exact consecutive classes needed to recover eligibility.

### 4.3 Daily Psychology Facts Engine

Inspired by the popular "Psychology Facts" app, ChemBuddy includes a curated database of **120+ evidence-based cognitive insights**. A new fact rotates daily at 8:00 PM IST using a prime-weighted calendar algorithm.



Figure 4.1: Home Dashboard showing the Claude-style greeting, attendance health ring (82.4%), daily psychology fact, and quick access tiles.

## 5. Smart Timetable & Attendance Intelligence Engine

The attendance module uses predictive mathematical equations to keep students compliant with the university 75.0% threshold.

### 5.1 The 75% Cutoff Mathematics

#### 1. Classes Needed to Recover 75%:

$$\text{Needed} = \max\left(0, \left\lceil 3 \times \frac{\text{Absent}}{\text{Present}} \right\rceil\right)$$

Indicates the exact consecutive classes required without a single absence to reach 75.0%.

#### 2. Safe Bunks Available:

$$\text{Can Skip} = \max\left(0, \left\lfloor \frac{\text{Present} - 3 \times \text{Absent}}{3} \right\rfloor\right)$$

Shows how many classes can be safely skipped while remaining strictly above 75.0%.

### 5.2 Attendance States

- **Present (✓):** Increments both Present and Total counts.
- **Absent (X):** Increments Total count only.
- **Postponed / Cancelled (||): Excluded from calculations.**  
Does not penalize your percentage when a professor is absent or a holiday occurs.
- **On Duty (OD):** Treated as Present for official university events.

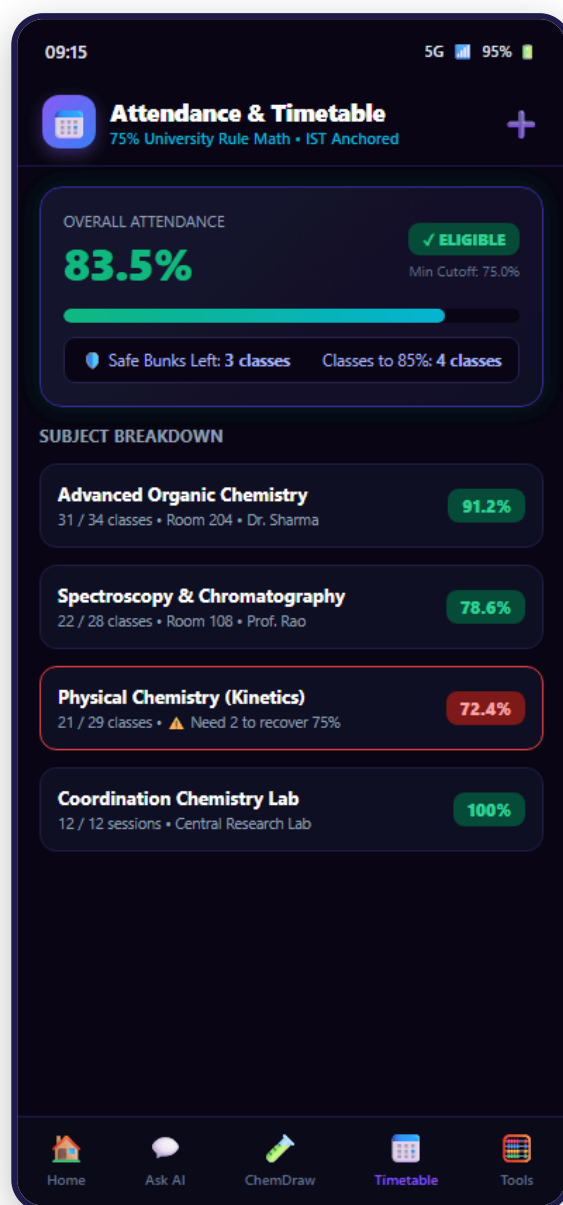


Figure 5.1: Timetable & Attendance Screen displaying the overall 83.5% meter, safe bunk calculation, and per-subject breakdown.

## 6. Notification System & 1-Tap Action Shade

ChemBuddy includes a specialized notification engine anchored strictly to the **Asia/Kolkata (IST)** timezone:

### 6.1 1-Tap Action Shade

When a class notification arrives 30 minutes before your lecture, you don't even need to open the app! Directly in the Android notification shade, tap:

- **Present ✓:** Instantly logs your attendance as present in the local database.
- **Absent X:** Instantly logs your attendance as absent.

### 6.2 Notification Channels

- **8:30 AM Timetable Briefing:** Daily morning overview of all scheduled lectures, room numbers, and attendance health.
- **30-Min Class Alerts:** Pre-class reminder with room number and current subject attendance.
- **8:00 PM Psychology Facts:** Daily cognitive insight to foster productive study habits.

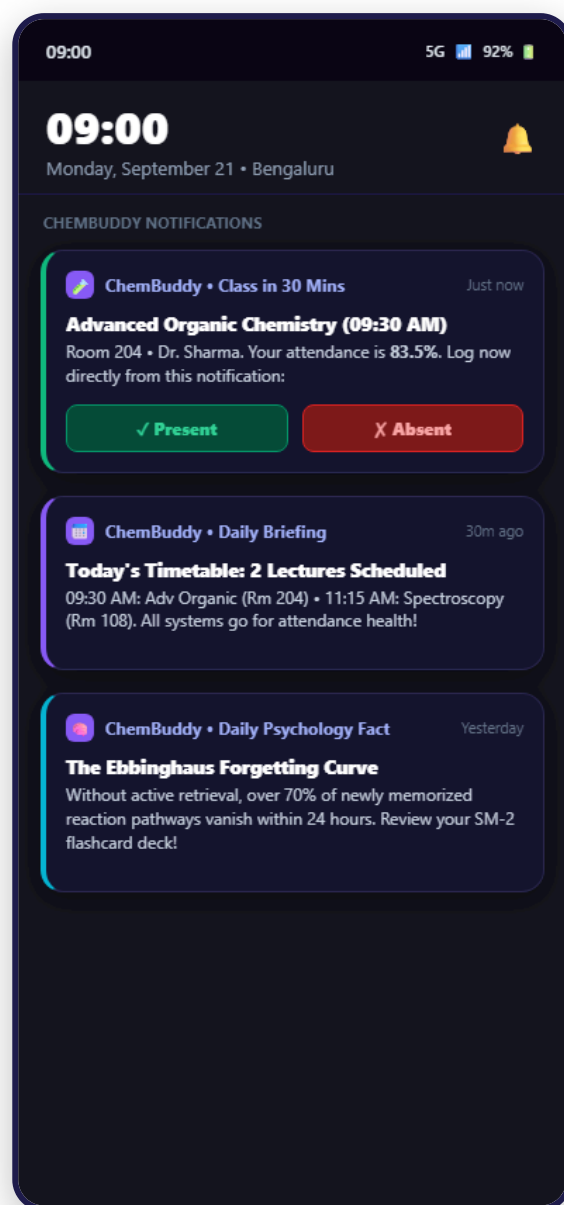


Figure 6.1: Android Notification Shade showing 1-tap Present/Absent action buttons, 8:30 AM briefing, and 8:00 PM psychology fact.

## 7. ChemDraw Pro & Ketcher Pro (118-Element Molecular Sketcher)

## 8. Reaction Product Prediction & 2D Vector SVG Generator

### 7.1 Molecular Sketcher Capabilities

- **Full 118-Element Periodic Table:** Searchable modal with standard CPK color coding.
- **Bond Tools:** Single, double, triple, wedge (stereochemical up), dash (stereochemical down), and curved electron arrows (  ).
- **Formal Charges:** +1, -1, +2, -2 with circled badges and   Molfile export.
- **Marquee Selection (  ) & 2D Clean-Up (  ): Standardizes bond lengths to 55px and optimizes angles.**
- **Multi-Reactant Mode:** When two disconnected molecules are drawn on canvas, ChemBuddy renders a cyan  symbol between them and outputs dot-separated SMILES (e.g. C1=CCC=C1.c1ccccc1).

### 8.1 Reaction Predictor & 2D Vector SVG

- **Gemini AI Internet Prediction:** Queries Google Gemini AI to predict the major product, formal reaction name, mechanism class, and curved electron-pushing arrows.
- **Publication-Quality 2D Vector SVG:** Replaced raw text boxes with crisp, scalable vector SVGs rendered directly in the prediction bottom sheet for Diels-Alder adducts, biaryls, and rings.
- **45+ Instant Offline Rules:** Zero-latency local fallback ensures predictions always succeed even without internet.

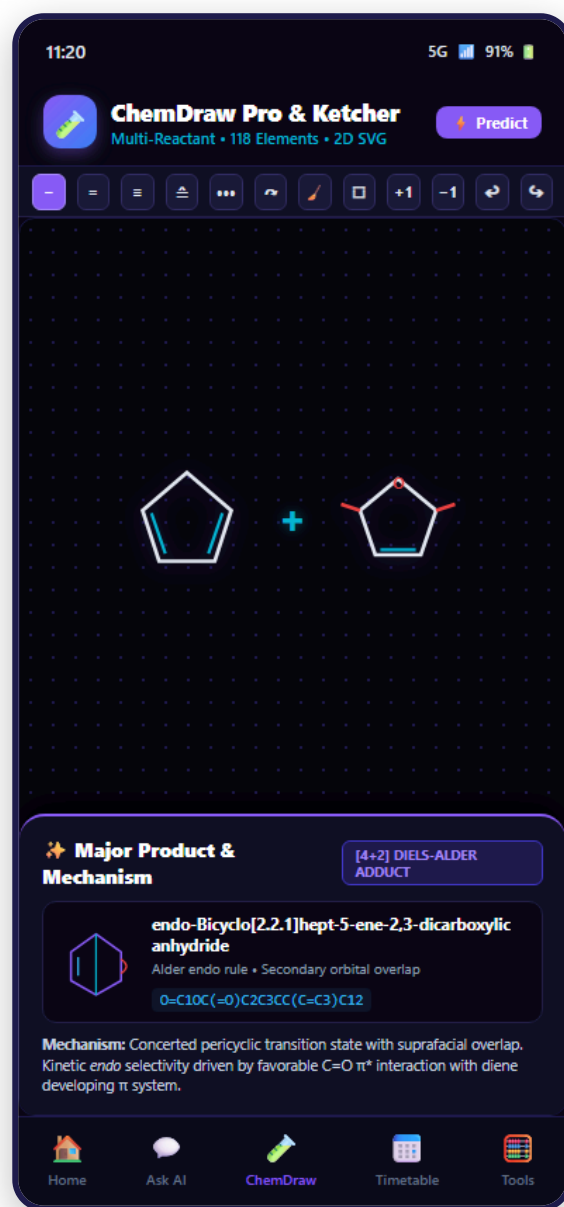


Figure 8.1: ChemDraw Pro canvas with Cyclopentadiene + Maleic Anhydride and the 2D Vector SVG prediction bottom sheet.

## 9. Spectroscopy & Chromatography Hub & Structure Solver

The **Spectroscopy Hub** covers 8 major analytical techniques and features an intelligent Structure Solver capable of identifying unknown compounds from spectroscopic data.

### 9.1 8 Analytical Techniques

- **RP-HPLC & GC:** Retention times, plate counts (N), resolution (Rs).
- **500 MHz  $^1\text{H}$  NMR &  $^{13}\text{C}$  DEPT:** Chemical shifts, multiplicity, coupling constants (J), DEPT-45/90/135.
- **FT-IR:** Characteristic functional group frequencies (carbonyls, O-H, N-H, triple bonds).
- **Mass Spectrometry:** Molecular ion peak ( $M^+$ ), base peak, McLafferty rearrangement.
- **UV-Vis:** Woodward-Fieser rules for conjugated dienes and enones.

### 9.2 Structure Solver Intelligence (52+ Molecules)

- **Double Bond Equivalent (DBE):** Automatically computes degrees of unsaturation from molecular formula.
- **Constitutional Isomer Ruling-Out:** Explains systematically why alternative isomers are excluded based on coupling patterns and chemical shifts.
- **Interactive Zoom (1.0x - 5.0x):** Smooth pan and zoom with haptic peak snapping.

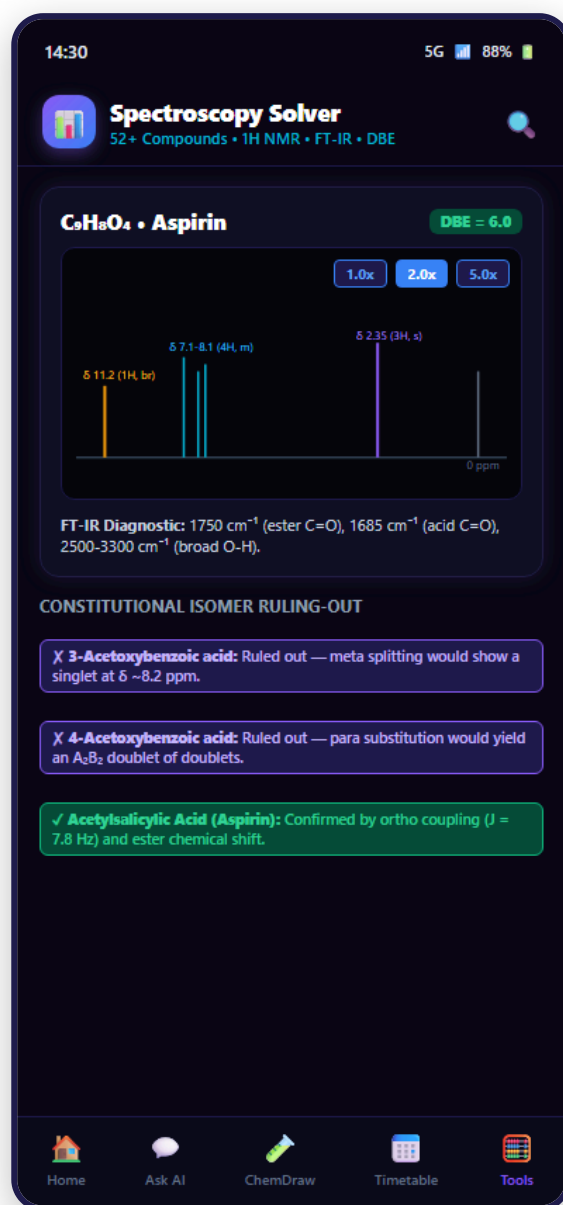


Figure 9.1: Spectroscopy Solver analyzing Aspirin ( $\text{C}_9\text{H}_8\text{O}_4$ , DBE=6.0) with zoomable  $^1\text{H}$  NMR spectrum and isomer exclusion rationale.

## 10. MSc Chemistry Toolkit & 16 Postgrad Calculators

ChemBuddy includes 16 specialized postgrad calculators organized into 6 practical laboratory domains:

- **Solutions & Concentrations:** Molar Mass, Molarity (g/L & M), Dilution ( $C_1 \cdot V_1 = C_2 \cdot V_2$ ).
- **Acid-Base & Buffers:** pH/pOH, Henderson-Hasselbalch, Buffer Formulation Assistant (Acetate, Phosphate).

- **Thermodynamics & Kinetics:** Gibbs Free Energy (  $\Delta G = \Delta H - T\Delta S$  ), Arrhenius Rate Law (Activation Energy  $E_a$  ).
- **Spectroscopy:** Beer-Lambert Law (  $A = \epsilon \cdot c \cdot l$  ), Photon Energy (  $E = h \cdot c / \lambda$  ).
- **Electrochemistry:** Nernst Equation (  $E = E^\circ - (RT/nF) \cdot \ln(Q)$  ), Standard Cell Potential.
- **Chromatography:** HPLC Calibration Curve (  $y = mx + c$  ), Theoretical Plates (  $N = 16 \cdot (t_R/W)^2$  ), Resolution (  $R_s$  ), Kovats Retention Index (  $I$  ).

## 11. Pericyclic Hub & Woodward-Hoffmann FMO Engine

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- **Frontier Molecular Orbital (FMO) Theory:** Visualizes HOMO and LUMO symmetry under thermal ( $\Delta$ ) and photochemical ( $h\nu$ ) conditions.
- **Electrocyclic Reactions:**  $4n$  systems (Thermal: Conrotatory, Photo: Disrotatory) and  $4n+2$  systems (Thermal: Disrotatory, Photo: Conrotatory).
- **Cycloadditions:**  $[4+2]$  Diels-Alder (Suprafacial-Suprafacial) and  $[2+2]$  Photocycloadditions.
- **Sigmatropic Rearrangements:**  $[1,3]$ ,  $[1,5]$ , and  $[3,3]$  Cope and Claisen rearrangements.

## 12. Smart Flashcards & Spaced Repetition (Active Recall)

The **Smart Flashcards** engine implements the scientifically proven **SuperMemo-2 (SM-2)** spaced repetition algorithm to guarantee long-term retention of reaction mechanisms, reagents, and spectroscopic values.

### 12.1 4-Point Confidence Ratings

After flipping a card to reveal the answer, rate your recall:

- **Again (1):** Interval resets to <1 day.
- **Hard (2):** Modest interval increase (~3 days).
- **Good (3):** Standard exponential increase (~6 days).
- **Easy (4):** Maximum interval increase (~12+ days).

### 12.2 Auto-Deck Generation

Tap "**Save as Flashcard**" on any response inside Ask ChemBuddy AI or import a syllabus PDF to automatically generate a tailored flashcard deck with questions and model answers.

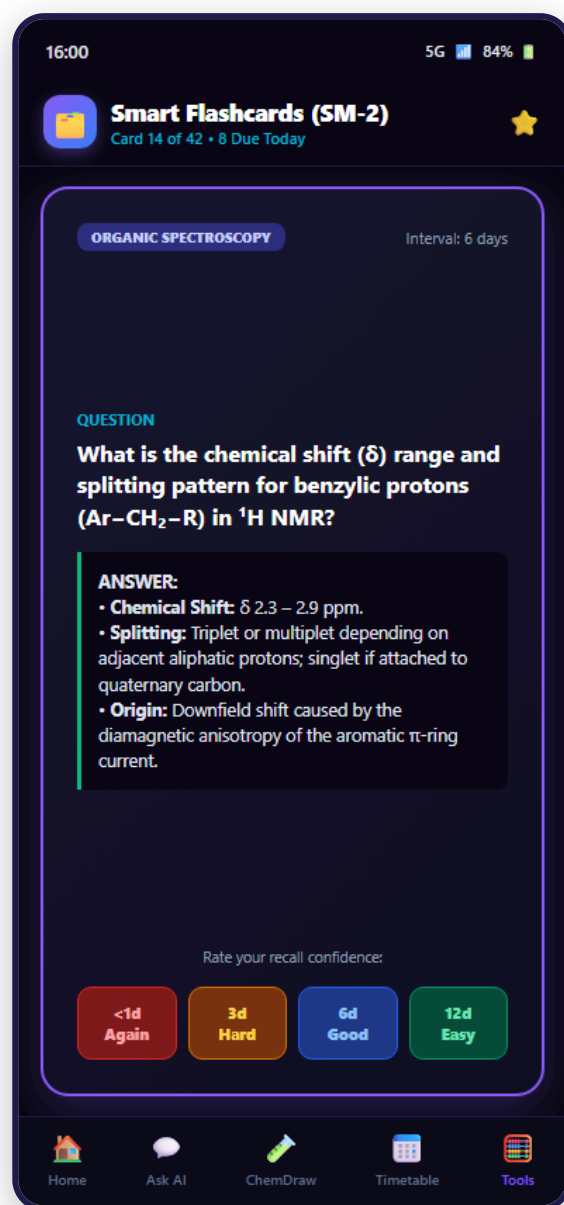


Figure 12.1: Flashcards active recall screen showing question card, diagnostic answer breakdown, and SM-2 rating buttons.

## 13. PDF Library, Study Hub & Strictly Grounded RAG Assistant

- **Offline PDF Viewer:** High-performance document reader with page jumping, keyword search, and zoom.
- **Local Text Extraction:** Extracts text locally on your device without uploading files to third-party servers.
- **Strictly Grounded RAG Chat:** In "From My PDF" mode, ChemBuddy's AI responses are strictly grounded in your uploaded documents. If the answer is not in the text, it explicitly states so, preventing hallucinations.

## 14. Exam Mode & BCU Model Answer Generator

Specifically calibrated to the examination patterns of Bengaluru City University (BCU), Central Universities, and state university MSc programs:

- **2-Mark Mode:** Generates crisp 2-to-3 sentence definitions, balanced equations, or IUPAC names for Section A questions.
- **5-Mark Mode:** Formats answers into the standard 5-part university rubric: Principle, Balanced Reaction, Stepwise Mechanism with Intermediates, Synthetic Applications, and Summary.
- **10-Mark Mode:** Comprehensive long-form essays complete with orbital transition state diagrams, stereochemical proofs, retrosynthetic analysis, and industrial examples.

## 15. Settings, Local Backup, Privacy & Offline Architecture

- **Notification Toggles:** Individual controls for Class Reminders, 8:30 AM Timetable Briefing, 8:00 PM Psychology Facts, and Exam Deadlines.
- **Instant Test Dispatch:** Tap "Send Test Notification" or "Send Psychology Fact" to verify notifications immediately.
- **Local JSON Export & Restore:** Export all profile records, attendance logs, timetable entries, and custom flashcards as an unencrypted portable JSON file. Restore seamlessly on any new device.
- **Zero Cloud Lock-in:** ChemBuddy never requires an account or cloud sign-in. Your data belongs exclusively to you.

## 16. Troubleshooting & Frequently Asked Questions (FAQ)

### Q1: Why didn't my 8:30 AM timetable notification sound?

**Answer:** On Android 12+, ensure that "Alarms & Reminders" permission is granted in Android System Settings. Additionally, set battery usage for ChemBuddy to "Unrestricted" so Android's Doze mode does not defer morning alarms.

### Q2: Should I keep my older app or install the new APK?

**Answer:** You must **uninstall / delete the older app** and install `ChemBuddy-v3.5.1-by-Prajwal-A-Kambar.apk`. The new APK contains the updated graph BFS/DFS SMILES generator, Gemini AI reaction prediction, and 2D vector SVG engine compiled into the native binary.

### Q3: How does ChemBuddy calculate attendance if a class is cancelled?

**Answer:** Mark the class as *Postponed / Cancelled (/)*. ChemBuddy excludes that class from both the numerator and denominator, ensuring your percentage is never unfairly penalized.

**Q4: Does Ask ChemBuddy work without internet?**

**Answer:** The core features (attendance, 16 calculators, spectroscopy solver, pericyclic rules, flashcards, and 45+ offline reaction predictions) work 100% offline. Online Gemini AI mode connects over the internet to generate dynamic mechanisms for unlisted or custom molecules.

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**ChemBuddy v3.5.1 Visual Operating Manual**

Engineered with ❤️ for MSc Chemistry Students worldwide by **Prajwal A Kambar**.